

Input and Output Devices

Lesson Objectives

By the end of this lesson, students should be able to:

Identify and explain different **types and functions of input and output (I/O) devices**.

- Understand the basics of **Human–Computer Interaction (HCI)** and its importance.
 - Describe **emerging I/O technologies** such as VR, AR, and biometric sensors.
 - Document and analyze I/O devices used in their personal environment.
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1 Introduction to I/O Devices

A **computer system** interacts with its environment through **input and output devices**.

- **Input devices** allow users to send data or commands *to* the computer.
- **Output devices** allow the computer to present processed data *to* the user.

Example:

When you type on a keyboard (input), the computer displays the text on a monitor (output).

2 Types of Input Devices

◆ a) Text Input Devices

Used to enter text, commands, and characters.

- **Keyboard** – standard input device for text.
- **Touchscreen keyboard** – found on phones and tablets.

◆ b) Pointing Devices

Used to control a pointer or select items on a screen.

- **Mouse** – most common pointing device.
- **Touchpad** – laptop input surface.
- **Stylus or Pen** – used for drawing or handwriting input on tablets.

◆ c) Scanning Devices

Capture images or data from physical documents.

- **Scanner** – converts paper documents to digital format.
- **Barcode Reader** – reads product codes.
- **QR Code Scanner** – common in mobile devices.

◆ d) Audio/Visual Input Devices

Used to record sound or images.

- **Microphone** – captures audio.
- **Webcam / Digital Camera** – captures images or video.
- **Headset with Mic** – combines audio input and output.

◆ e) Sensor-Based Input Devices

Capture environmental or physical data.

- **Fingerprint Scanner** – captures biometric patterns.
- **Motion Sensor** – detects movement.
- **Temperature Sensor** – used in IoT systems.

3 Types of Output Devices

◆ a) Visual Output

Displays images, video, or text.

- **Monitor / Display Screen** – main output for computers.
- **Projector** – displays large-scale visuals for audiences.
- **VR Headset** – immerses users in 3D environments.

◆ b) Audio Output

Produces sound.

- **Speakers** – output audio from applications or media.
- **Headphones / Earbuds** – personal audio output devices.

◆ c) Hardcopy Output

Provides permanent physical copies.

- **Printer** – prints documents or images on paper.
- **Plotter** – prints vector-based drawings (used in engineering).

◆ d) Haptic Output

Uses touch feedback to communicate.

- **Vibration motors** – found in game controllers and smartphones.
- **Force feedback devices** – used in simulation systems.

4 Functions of I/O Devices

Function	Input Device Example	Output Device Example
Entering text	Keyboard	—
Drawing	Stylus	—
Viewing results	—	Monitor
Hearing sounds	—	Speakers
Recording video	Webcam	—
Printing document	—	Printer

💡 **Note:** Some devices serve *both* as input and output — for example, **touchscreens**, **headsets**, and **VR devices**.

5 Human–Computer Interaction (HCI) Overview

◆ Definition

HCI is the study and design of how humans interact with computers. It aims to make systems **usable, efficient, and enjoyable**.

◆ **Key Goals of HCI**

- Improve **usability** (ease of use).
- Enhance **user experience (UX)**.
- Support **accessibility** for all users.
- Design **intuitive interfaces**.

◆ **HCI Design Principles**

Principle	Description
Feedback	The system should respond visibly to user actions.
Consistency	Similar operations should have consistent results.
Affordance	Interface should suggest its function (e.g., buttons look clickable).
Simplicity	Reduce unnecessary complexity in design.

◆ **Examples of HCI in Action**

- Touch-based smartphone interfaces.
- Voice assistants (Alexa, Siri, Google Assistant).
- Gesture-based VR controls.

6 Emerging I/O Devices

Technology is evolving beyond traditional I/O tools. Here are some examples of **modern devices** shaping new ways of interaction:

Device Type	Description	Example Use
Virtual Reality (VR) Headset	Immerses user in simulated 3D world	Gaming, medical simulation
Augmented Reality (AR) Glasses	Overlays digital info on real world	Navigation, education
Biometric Sensors	Reads physical features	Unlock devices via fingerprint/face

Brain–Computer Interface (BCI)	Uses brain signals to control systems	Neuroprosthetics, research
Gesture Control Systems	Recognizes hand/body movements	Smart TVs, AR/VR environments
Voice Recognition Devices	Understands speech commands	Smart speakers (Alexa, Google Home)

Activity: Document Your I/O Devices

Objective:

Identify and analyze the input and output devices you use daily.

Instructions:

1. Look around your home or study setup.
2. List **at least five input devices** and **five output devices**.
3. For each device, document:
 - Device name
 - Type (input/output/both)
 - Function or purpose
 - Connection type (wired, wireless, Bluetooth, etc.)

Sample Table:

Device Name	Type	Function	Connection
Keyboard	Input	Enter text	USB
Mouse	Input	Point/select	Wireless
Monitor	Output	Display visuals	HDMI
Headset	Input/Output	Audio input/output	Bluetooth
Printer	Output	Produce hard copy	Wi-Fi



Git Task

1. Create a markdown file named:
2. week5_io_devices.md
3. Include your completed table and a short paragraph summarizing:
 - How I/O devices help you interact with your computer.

- Which devices you use most frequently and why.
 - 4. Add and commit to your repository:
 - 5. `git add week5_io_devices.md`
 - 6. `git commit -m "Added Week 5 I/O devices documentation"`
 - 7. `git push origin main`
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